

🕒 Lumen — the web, readable for everyone.

An edge-native accessibility agent built on Tencent EdgeOne Makers · **Live:**
lumen.edgeone.cool · **Code:** github.com/PranavAchar01/lumen · Built by
Pranav Achar

AGENT FORGE MINI HACKATHON

AI BUILDERS × TENCENT EDGEONE

AI ENGINEER WORLD'S FAIR, SF





95% of the Web Fails Accessibility

The **WebAIM Million study** found critical failures on ~95% of the top 1M homepages — missing alt text, broken heading structure, unlabeled buttons, walls of clutter.

285M+ people with visual impairment — plus millions with cognitive or reading disabilities — experience most of the web as exhausting or unusable.

Missing Alt Text

Images invisible to screen readers

Broken Headings

No navigable page structure

Unlabeled Buttons

Actions impossible to identify

Opt-In Overlays

Only fix sites that choose to act



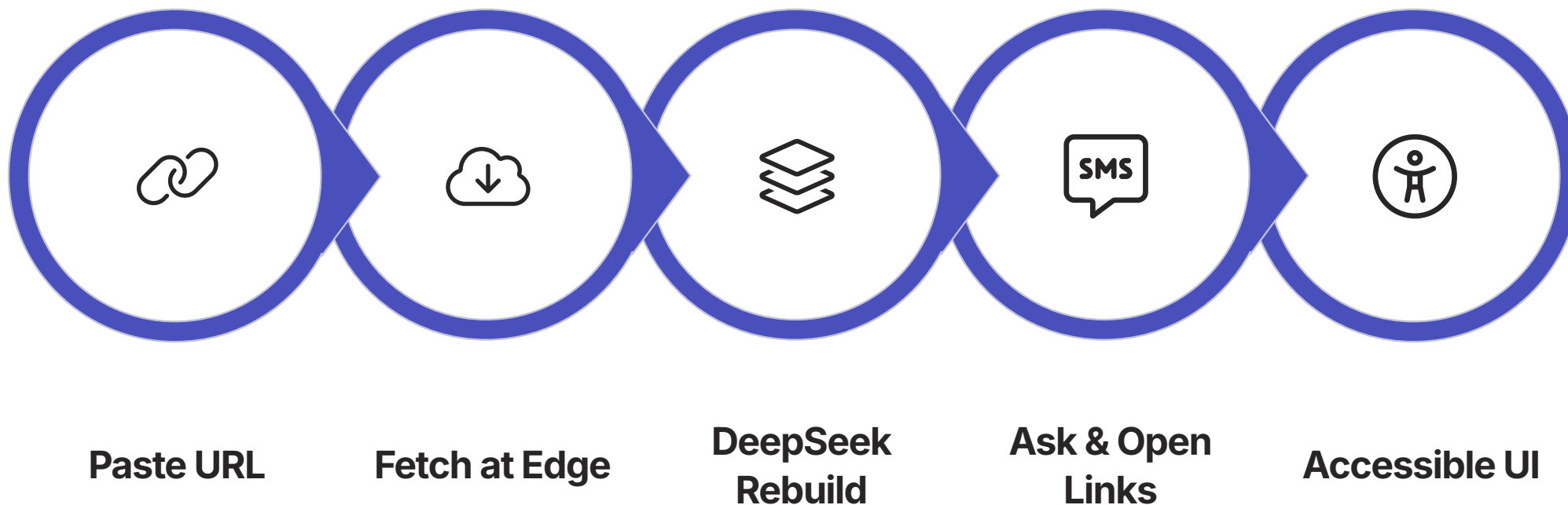
Today's accessibility overlays only work on sites that opt in. The reader is left powerless on the other 95%.

The Insight

Stop waiting for the web to fix itself.

Give the **reader** an agent that repairs any page — on demand, at the edge, for free.

What Lumen Does



Every step happens at the nearest of EdgeOne's 3,200+ edge nodes — no origin server, no API keys. The rebuilt view includes a plain-language summary, navigable page map, key actions, and flagged accessibility traps. Ask anything and the agent can **open links autonomously** to find your answer.

Live Demo — Try It Now

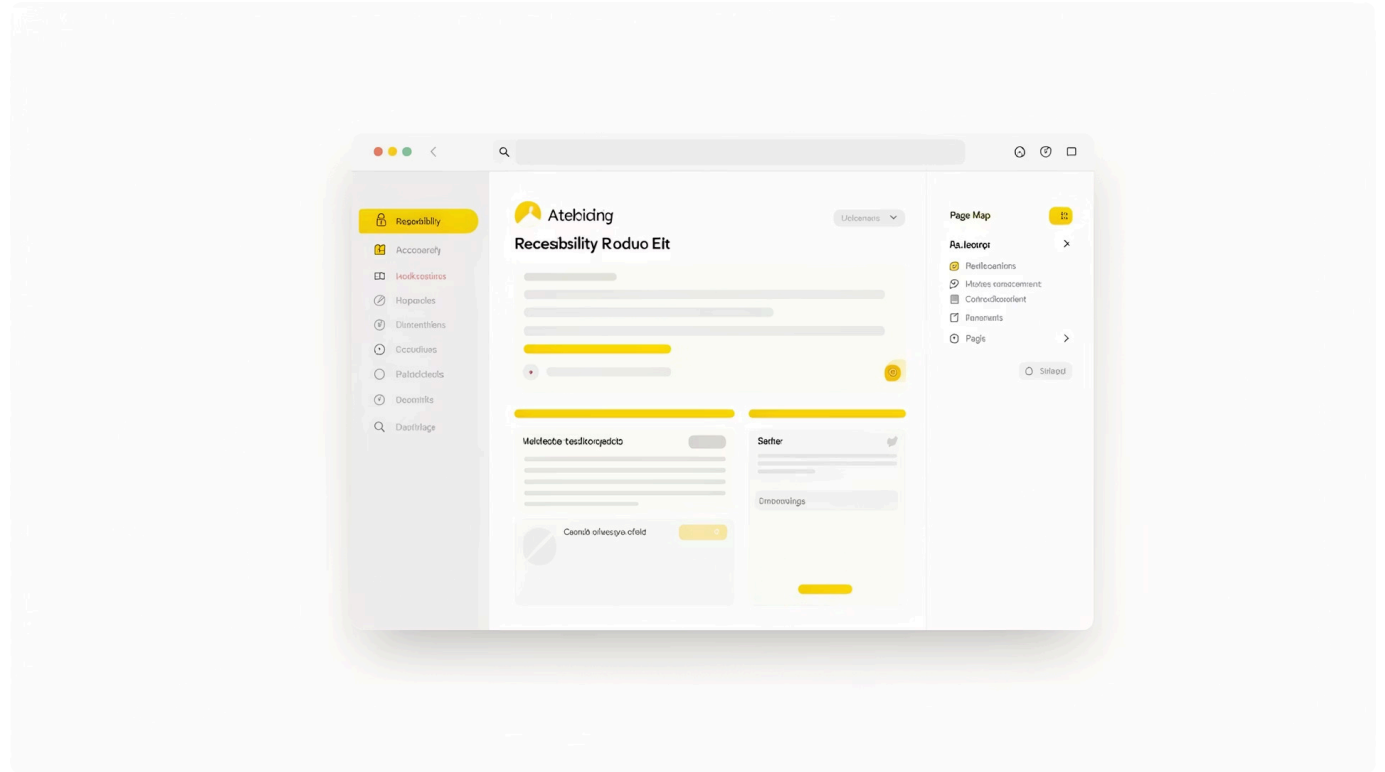
lumen.edgeone.cool → click "W3C: Intro to Accessibility" → **14 seconds** → clean reading view with page map and actions.

Ask: *"What are the three examples of accessibility this page gives?"* → exact, faithful answer every time.

Shareable deep links:

lumen.edgeone.cool/?u=<any page>

- ✅ Deployed, working, and verifiable live — right now.



Architecture: 100% on EdgeOne



EdgeOne Pages

Static UI on global CDN — zero origin server



Edge Functions

Entire agent: fetch → extract → rewrite → answer



Edge AI · DeepSeek

Built-in via `AI.chatCompletions`, auto-fallback across 3 model variants



EdgeOne KV

24h cache — one rebuild serves every reader worldwide instantly



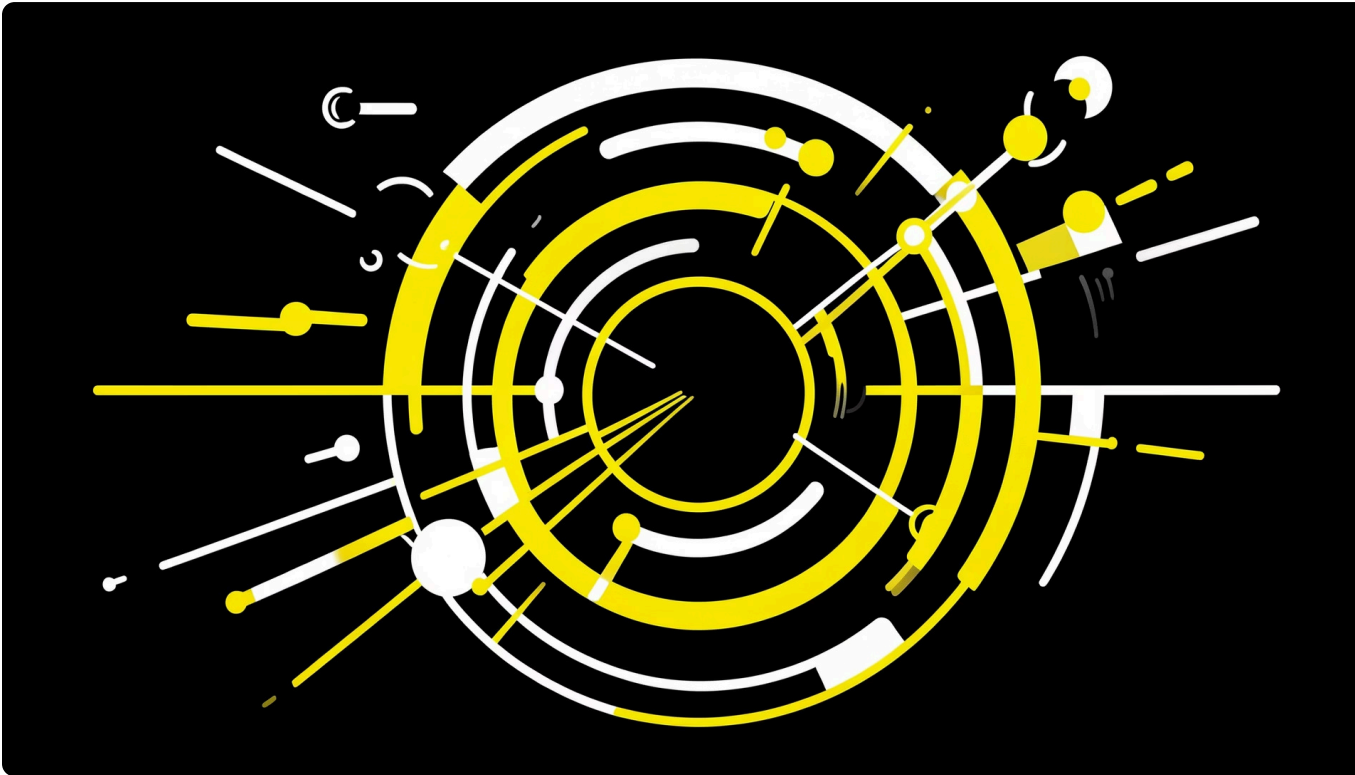
Geo Metadata

`request.eo` surfaces which node served you



EdgeOne CLI

Headless deploys — one command to ship



Latency IS Accessibility

Screen-reader users consume pages **serially** — every extra second of wait costs them disproportionately more than a sighted user skimming visually.

Nearest-Node Fetching

3,200+ nodes — always the shortest path to the content

Shared KV Cache

One reader's rebuilt page = every reader's instant page, worldwide

Zero Infrastructure

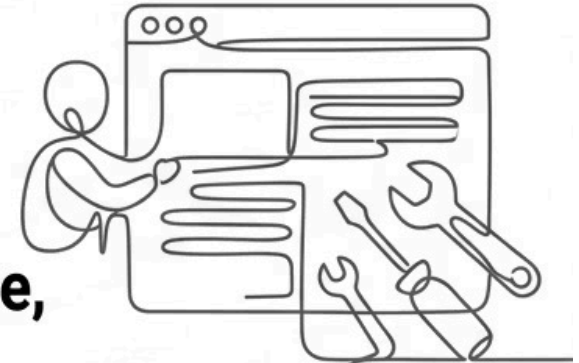
Nothing to run, scale, or maintain

Judging Criteria Scorecard



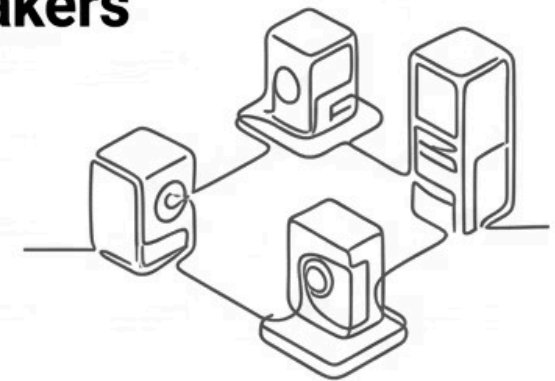
COMPLETENESS:
Deployed live
MVP, public repo,
end-to-end
verified.

INNOVATION:
Inverts
accessibility-
overlay model,
repairs ANY page,
visible agentic
tool use.



REAL-LIFE PROBLEM:
95% of web fails
accessibility,
285M+ affected,
ADA/EAA legal
pressure growing.

SPONSORED PRODUCT USAGE:
Six EdgeOne Makers
products in
production,
platform IS the
architecture.



Built in one afternoon. Deployed in one command. **Readable by everyone.**